Instructor guide / consolidated revision

Classical ciphers • Mathematics master’s students new to cryptography • 75 teaching slides + 13 appendix pages • 180 minutes including two breaks

Pacing plan — instructor use only

Elapsed time	Slides	Focus
00–16	1–11	Communication problem, terminology, symmetric keys, mathematical model
16–25	12–18	Worked shift; stop the search once key 3 gives HELLOWORLD
25–30	19–22	Define the chosen permutation σ and its inverse; work the table
30–60	23–32	Five representative recovery stages, partial plaintexts, and guided practice
60–70	33–33	Break
70–120	34–50	Worked encryption; visual repetitions; IC intuition; coloured columns; key recovery
120–130	51–51	Break
130–158	52–66	Square construction, preparation, geometric rules, and exercise
158–172	67–73	Forward and inverse permutations, calculation, and exercise
172–180	74–75	Compare the constructions and connect to digital messages

Use the shorter explanations to make room for understanding

The lecture no longer displays a timetable or ends with the removed reflection and reading slides. Keep the pacing here. The smaller number of repeated trial-and-error slides gives time to annotate one partial key on the board, let students predict the next substitution, and revisit any unclear step.

Lead with the concrete examples and diagrams. Introduce notation after students understand the operation it describes. Optional IC derivations and chi-squared details are in the appendix; they are supporting material rather than required steps in the main presentation.

Foundations, shift, and substitution

A clear opening

The redesigned cover retains only the lecture title, series name, and introductory subtitle. Begin with Alice, Bob, and Eve and an ordinary plaintext message. Establish plaintext, ciphertext, algorithm, and key before formal notation. Today’s focus is confidentiality; assume a shared secret and postpone key distribution.

Introduce plaintext, ciphertext, and key spaces as sets. E_k maps plaintext to ciphertext; D_k reverses the chosen transformation. $D_k(E_k(p)) = p$ is a correctness condition, not a security claim.

Alphabet and indexing conventions

Use A = 0 through Z = 25 and representatives 0–25 for residues modulo 26. Message positions start at 0 in the shift and Vigenère sections. Permutation tables later use positions 1 through m explicitly. Removing spaces and punctuation is preprocessing and can discard information.

Shift: calculate, then recognize the answer

With key 2, C (2) → E (4), R (17) → T (19), Y (24) → A (0), and P (15) → R (17). The complete CRYPTOGRAPHYISFUN result is ETARVQITCRJAKUHWP. Reverse A → Y by subtracting 2 modulo 26.

For intercepted KHOORZRUOG, show trials 1, 2, and 3 only. Key 3 produces HELLOWORLD, so stop the worked search there. The mathematical key space still contains 26 candidates; there is no need to display a subsequent key after this plausible solution.

The exercise PHHWPHDIWHUFODVV decrypts to MEETMEAFTERCLASS at key 3. Known plaintext M → P gives $(15 - 12) \bmod 26 = 3$. Very short strings can have several plausible readings; this example is intentionally recognizable.

Explain σ before using it

S_{26} is the set of all alphabet permutations. The chosen secret key σ is one particular member of that set: it assigns each plaintext symbol exactly one distinct ciphertext symbol. Its inverse σ^{-1} follows the same table in reverse for decryption. The inverse identity explains why the original symbol is recovered.

The illustrative QWERTY alphabet maps BECAUSE to WTEQXLT. To reverse W, locate it in the output row and read B above it. The later 168-letter attack uses a different unknown key. There are 26! keys, but equality patterns and counts survive; key-space size alone does not establish security.

Frequency analysis / five representative stages

Slides 23–32 retain the original 168-letter message while combining repetitive guesses. Ciphertext is uppercase; partial plaintext is lowercase; a middle dot marks an unknown letter. Work one or two representative hypotheses carefully, then group the similar continuations.

Stage	Main reasoning
1. Count	Z occurs 20/168 times. Test Z → e while leaving other common symbols open.
2. Pair clues	ZW occurs 4 times, WZ zero times, RW twice, NZ three times, and ZN once. Together with injectivity and language clues, test W → d, R → n, and N → h.
3. Partial text	RZCRWNZ gives ne·ndhe, suggesting C → a. RNM gives nh· and CM gives a·; test M → i. Apply the resulting key to the whole text.
4. Word patterns	JN suggests th, the emerging context suggests o, and MD suggests is. HNCMF then suggests chair. Group these similar tests rather than repeating the complete procedure for every letter.
5. Complete	Use familiar fragments such as our friend, from Paris, looking, examined, and evaporation to summarize the remaining recovery. Check all proposed mappings globally.

Keep evidence and logical constraints distinct

Choosing Z → e is a statistical guess. Excluding R → e after retaining Z → e uses the bijection constraint. Other word completions remain hypotheses until they are consistent with the whole ciphertext. Do not treat a missing pair in a short text as proof of impossibility.

The verified ciphertext has one ZN occurrence. The original reference says zero; the deck uses the recomputed count. This correction does not require spending lecture time reconstructing the old error.

Reveal the full message after a visible partial solution

Our friend from Paris examined his empty glass with surprise, as if evaporation had taken place while he wasn’t looking. I poured some more wine and he settled back in his chair, face tilted up towards the sun.

Exactly 23 plaintext letters occur. J, Q, and Z are absent; 3! = 6 full substitution keys can extend the recovered partial key. The message can be readable even when the complete key is not uniquely determined.

Vigenère / explain the attack through examples

First work the encryption

Repeat LUCK below CRYPTOGRAPHY. The first six sums produce NLAZEI; the full result is NLAZEIIBLJJI. R at positions 1 and 7 encrypts to L and B because its aligned key letters differ. Decryption subtracts the same aligned residue.

See the repetition before calculating a period

In THESUNANDTHEMANINTHEMOON with KING, highlight THE at positions 9 and 17, the aligned key letters, and the two BUK outputs. The marked distance is 8, divisible by period 4. THE at position 0 has a different alignment.

The longer attack uses one 1,045-letter ciphertext, printed in Appendix A1–A2. FVR starts at positions 97, 147, 552, and 897; distances from the first occurrence are 50, 455, 800. Their gcd is 5. These are candidate-period clues, not a guarantee.

Define index of coincidence before abbreviating it

Index of coincidence (IC) is the probability that two randomly chosen, distinct positions contain the same letter. In AAB, the three unordered pairs are AA, AB, and AB: one matches, giving IC = 1/3. Ordered counting gives the same rate; the longer derivation belongs to the appendix.

For longer texts, uniform independent letters have expected IC about 0.0385; ordinary English prose often has a value near 0.065. Uneven letter frequencies cause more matching pairs. IC measures this match rate, not general cryptographic security. Show the exact count formula only after this intuition.

Use the coloured columns to explain the check

Lay out the ciphertext in rows of five and colour each column consistently. If the period is 5, each column shares one shift. The first five rows are ELQVV / DXAFL / ZJESP / CIFKE / TXUBT. Read down a column, not across it.

Period 1 gives IC 0.0429. At period 5, the five column values are 0.0646, 0.0679, 0.0638, 0.0679, 0.0699. Each column has 209 letters. A period of 10 can also look plausible because it divides the same aligned shifts into smaller samples; avoid claiming a unique answer from IC alone.

Recover one shift visually, then summarize

The first column contains P 27 times. If P represents shifted E, the candidate shift is (15 − 4) mod 26 = 11, or L. Compare the whole trial-decrypted histogram against the English model under shifts 0 and 11. The column itself is every fifth plaintext letter and will not form continuous words.

Summarize the remaining shifts as 11, 4, 12, 14, 13 → LEMON. Decrypting the complete interleaved message gives THEHISTORYOFSECRETWRITING… . Appendix A5–A6 contains the complete prose. Detailed numerical scoring remains optional appendix material.

Playfair and transposition

Playfair: build and prepare before transforming

Construct PLAYF / IREXM / BCDGH / KNOQS / TUVWZ from PLAYFAIR EXAMPLE. The two-stage diagram separates keyword placement from completion. I and J share a cell. BALLOON scans as BA, LX, LO, ON; reconsider the second L after inserting the filler.

Use X as the default filler and Q when the current letter is X. Thus X becomes XQ and XX becomes XQ XQ. This explicit convention handles the repeated-X edge case; other implementations may choose a different alternate.

Work the three geometric rules

Same row: PL → LA; wrapping FP → PL. Same column: PI → IB; wrapping TP → PI. Rectangle: H at (3,5) becomes B at (3,1), and I at (2,1) becomes M at (2,5), so HI → BM. Grid shading identifies the input pair.

The first gold-example pairs are HI→BM by rectangle, DE→OD by column, TH→ZB by rectangle, and EG→XD by rectangle. Work these before revealing the complete prepared and encrypted rows.

Playfair exercise and interpretation

RO→EN by rectangle; NC→CR by column-up; FL→YP by row-left; WI→TX by rectangle. The prepared result is ENCRYPTX; ENCRYPT is the intended reading after interpreting padding. AXA and AA both prepare to AXAX, so preparation can be many-to-one. Do not automatically remove every X or assume exact I/J spelling is recoverable.

Transposition: direction before calculation

Encryption reads input positions [3, 1, 4, 6, 2, 5]; decryption reads the inverse [2, 5, 1, 3, 6, 4]. ATTACK→TAAKTC; ATDAWN→DAANTW. The exercise gives DEFEND→FDEDEN→DEFEND. There are 6! = 720 keys at this block size, and labeled letter counts are unchanged.

Construct the inverse by asking where each original input position went. Composing the permutation and its inverse proves correctness. For incomplete blocks, retain the true message length separately from X padding.

Optional depth, reading, and reproducibility

Use the appendix selectively

The complete Vigenère ciphertext, count table, frequency model, and recovered prose support independent study. The longer IC derivation, sample/probability identities, chi-squared score formula, contribution calculations, and candidate-score tables provide optional mathematical depth. The main lecture supplies the intuition and the essential IC formula first.

Language scores rank hypotheses and can fail on short or unusual text. Re-encryption checks consistency with a candidate transformation; it cannot by itself establish that the key is correct. Coherent full-message plaintext supplies further evidence.

Reading and provenance — retained here rather than on a lecture slide

Stinson & Paterson, *Cryptography: Theory and Practice*, 4th edition, Chapter 1, provides core reading. Section numbering differs across editions. The supplied *Lecture02_Classical_Ciphers.pdf*, Wu Hongjun, supplies the classical-cipher scope, 168-letter substitution example, and standard Playfair illustration.

Historical reading: Kahn, *The Codebreakers* (revised edition, 1996); Kerckhoffs, *La Cryptographie Militaire* (1883); Kasiski (1863). The original uploaded PDF is cited but not redistributed in the source bundle.

This consolidated revision

Applies the recorded cover redesign and slide removals, stops the worked shift search at key 3, defines σ explicitly, groups repeated substitution reasoning into five stages, and makes Kasiski/IC visual and example-led. It retains the same five cipher constructions and selected exercises. Page numbers in the final PDF reflect the removals and regrouping.

Edit, build, and inspect

Requirements: Python 3 with ReportLab and PyMuPDF; DejaVu Sans and Sans Mono font files; pdflatex with amsmath, amssymb, Latin Modern, xcolor, and geometry. Equations are placed as vector PDF content; LaTeX shell escape is disabled.

Run python build_lecture.py, python build_guide.py, then python validate.py from lecture01. Edit slide_content.py for content, cipher_examples.py for examples, and build_lecture.py for layout. The manifest, numerical data, PDFs, and validation report are generated outputs. Inspect rendered pages after edits as well as running the checks.

The source bundle includes editable files and supporting data. Previous published versions remain accessible through Git history.